Priorities of woody species trait-climate associations at continental scale

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Abstract/ Summary:

Plant functional traits underpin ecological strategies and determine ecosystem responses to climate change, yet the key climate drivers of traits remain unresolved at continental scale. Key questions are whether across large gradients species traits are driven more strongly by temperature or precipitation—and by annual or by growing season mean climates. Across 16 ecosystems across the continental USA and northern Mexico, we measured 328 species-site combinations, including 246 unique woody species, for 59 traits relating to hydraulic and photosynthetic physiology, leaf and wood structure and anatomy, and nutrient and isotope concentrations using standard protocols, and tested associations with the macroclimate of both the species' natural distributions and of the sampling sites, i.e., growing season and mean annual temperature, precipitation, potential evapotranspiration, and aridity index. Across species-site combinations, ninety-eight percent of traits were associated with one or more climate variables. More traits were correlated, and showed stronger correlations, with mean and growing season precipitation than temperature, and more trait variation was explained by precipitation than temperature in multiple regressions. Potential evapotranspiration and aridity index were also strong trait predictors. Seasonal and annual means did not differ in the strength of trait correlations for precipitation, whereas growing season temperature was a stronger predictor of traits than mean annual temperature. Our findings highlight the importance of mean annual precipitation a driver of the distribution of plant traits, indicating the strong sensitivity of species to ongoing shifts in moisture availability at continental scale.

Introduction

Climate plays a critical role in shaping species adaptation and natural distributions. The global diversity of plants can be considered in terms of functional traits—i.e., the morphological, phenological and physiological characteristics that determine reproduction, growth, and survival strategies[1], which are frequently aligned with the environment, including macro- and microclimate. Thus, climate shapes the assembly and evolution of species' traits, and, conversely, a species' traits determine the climates in which the species can recruit, survive, compete and regenerate[2–6]. Indeed, plant trait-climate associations have been studied at least since Von Humboldt (1807) described how vegetation becomes shorter at higher elevations and latitudes, and a wealth of classical studies have highlighted many factors that underlie the dramatic patterns in plant adaptation and ecological strategies, with temperature and precipitation arguably the most analyzed variables[7–14].

Temperature and moisture influence tissue temperature and hydration, and thereby impact gene expression, photosynthetic capacity (photosynthetic rate, respiration), hydraulic function, and growth (tissue expansion, relative growth rates)[15–24]. Further, exposure to either extremely high or low temperatures or moisture supply can damage or kill plants[23,25–27]. Plant traits may contribute to stress-tolerance, through "resistance" mechanisms, i.e., enabling the maintenance of function during stress, and/or through "avoidance" mechanisms, i.e., reducing function during stress and mitigating the loss with rapid growth during periods of favorable climate[6,28,29]. Thus, within and across species, numerous traits correlate with temperature and/or precipitation, including leaf size, leaf mass per area, leaf nitrogen and phosphorous concentrations, photosynthetic rate, wood density, seed mass, plant height, and water use efficiency[11,16,30–35,35–40].

While plants can respond to given climate variables semi-independently, ultimately performance would depend on their combined influence[6,39,41]. Thus, given natural covariation of environmental factors, plant species may be adapted, assembled and/or adjusted plastically, according to the combinations of macro- and microclimate variables that they experience, in addition to other abiotic and biotic factors[42–44]. A topic of strong current interest is parsing out key climatic determinants of biogeographic trait distributions, which would reveal how specific pressures structure plant functional strategies, enable more accurate trait-based predictions, and advance the understanding of adaptation limits [45]. Thus, many studies have focused on the associations of plant traits with the mean climate of species' or ecotypes' native ranges, at a range of scales, across closely-related species or ecotypes[46–49], or across species from diverse ecosystems, within or across biomes[6,11,40].

Yet, few studies have tested the relative numbers and strengths of the associations of traits with climate variables, i.e., the "priorities" of trait-climate associations. Indeed, the importance of temperature versus precipitation as a driver of species' trait associations across large spatial extents has become a topic

of debate. A highly-cited study reported that annual temperature is a better predictor than precipitation of plant traits globally, considering 21 traits from a compiled trait database for 447,961 site-species combinations [11]. However, ecological patterns at continental scales may differ from those at global scales (Table 1), given contrasting ranges of variation in traits and in climate variables, which are influenced by multiple interacting factors. For example, at a global scale, trait–climate associations correspond to broad latitudinal gradients in which temperature and moisture are positively associated from poles to tropics, and represent high species turnover and phylogenetic heterogeneity, while within continents, aridity gradients may be stronger due to positive association of dry and warm environments, and trait-climate associations may be strengthened by shared floristic histories and more limited species pools. A focus on the adaptation and assembly of ecosystems at continental scale has been established as a Grand Challenge for predicting and mitigating the impacts of climatic shifts[50], as such systems are models for understanding shifts under hotter, drier future climates[51]. One study at continental scale in Australia found mean annual precipitation was a strong predictor of eight key functional traits[52].

Here we provide a strong test of the central outstanding question of the priorities of trait-climate associations for woody species across ecosystems of the continental USA and Mexico. We focused on four climate variables commonly considered in ecology— temperature, precipitation, potential evapotranspiration, and aridity index (i.e., the ratio of precipitation to potential evapotranspiration). We considered 59 traits including hydraulic, morphological, structural, photosynthesis and compositional traits, expected to contribute to stress resistance and/or stress avoidance (see hypotheses and references in Table S1), We focused on 328 species-site combinations including 246 unique species, sampled in 16 diverse woody-dominated ecosystems representative of larger regions (Fig. 1a; Table 1; Tables S2 and S3). We overcame several limitations of previous studies that would weaken trait-climate associations at large spatial scales. Previous studies have analyzed databases compiled for plants measured at many different locations and times, sometimes using differing methods, with a focus on few traits, especially relating to fast vs. slow growth ("leaf economic spectrum" traits) or plant size, which globally constitute orthogonal axes, often independent of climate[11,39,42,52]. These methodological considerations are among a larger set of reasons that would lead to weakened trait-climate correlations, also including trait-climate mismatch, trait multifunctionality, many-to-one trait mapping, disequilibrium effects, the influence of microclimate, intraspecific trait and climate variation, and climate sampling bias[6,54] .We focused on measurements for plants coexisting in ecosystems across the gradient, and measured a wide range of traits using standard methods, as the incorporation of additional mechanistic traits (including hydraulic and compositional traits) can increase power for resolving trait-climate relationships, especially when common protocols are applied to plants grown together in one or few common environments[6,44,55].

We developed the first continental-scale analysis of the priorities of trait-climate relationships across the United States of America and northern Mexico. We tested association of traits with the mean climate of species' native ranges, focusing on four key climatic variables known to shape plant and community distributions and diversity and community, considering annual and growing season means: growing season and mean annual temperature (GST and MAT, respectively), growing season and mean annual precipitation (GSP and MAP), growing season and mean annual potential evapotranspiration (GSPET and MAPET), and growing season and mean annual aridity index (GSAI and MAAI) (Fig. 1b). First, we hypothesized that across these sites, traits would correlate most closely with the aridity gradient, and thus, more strongly with precipitation, PET and aridity index, than with temperature. Second, we hypothesized that traits would correlate more closely with growing season than annual climate variables, under the expectation that traits would have strongest influence during the active growth period[56–58].

Methods

Data collection

Our study focused on data from the CommuniTraits Database (Medeiros et al., *to be submitted*). This database includes traits measured with standard methodology on 19 sites including 12 ecosystem types distributed across diverse climates throughout the United States: eight sites of the California floristic province that include alpine, desert, coastal sage scrub, shrubland, chaparral, montane wet forest, mixed riparian woodland and mixed evergreen-deciduous forest ecosystems; nine additional woody-dominated temperate evergreen, deciduous and mixed evergreen-deciduous forests from East to West coasts, and Hawaiian montane wet and lowland dry forests. For this comparative study, we focused on woody species and thus excluded the species of an alpine site dominated by herbaceous species, and given our focus on a continental gradient, we excluded data from two tropical Hawaii forests. Permits were obtained for work in the reserves through direct communication with the reserve directors for the sites part of the University of California Natural Reserve System (UCNRS), the National Ecological and Observatory Network (NEON), and the Forest Global Earth Observatory (ForestGEO). For the Yosemite Forest Dynamics Plot and the Great Smoky Mountains National Park, the permits were obtained through the United States Department of the Interior National Park Service (Permits #YOSE-2017-SCI-0009 and #GRSM-2023-SCI-2199, respectively).

Field sampling and trait measurements

Our database included the biomass-dominant and other common species at each site, selected based on guidance from reserve managers and, when available, using forest census data. The species included in this study are taxonomically diverse (Table S3), including 246 unique species from 79 families, with 328

153 species-site combinations across the 16 sites. At each site, we sampled 3–5 individuals per species, and 14-
154 26 species per site. Individual trees and shrubs were sampled across the landscape and we avoided sampling
155 adjacent individuals of the same species. We collected sun-exposed leaves from non-epicormic branches,
156 with no signs of damage or herbivory using aerial lifts, pole pruners or a slingshot. Branches were
157 transported to the lab in dark plastic bags with moist paper and rehydrated overnight in a saturated
158 atmosphere before harvesting current-year grown, fully expanded leaves for subsequent analyses. For
159 compound-leafed species, whole leaves were used. We measured 59 traits using standard protocols (traits
160 listed in Table S1; detailed methods described in Methods S1). For epidermal morphology, we measured
161 one leaf per individual (epidermal cell areas, stomatal size, density, index, maximum anatomical
162 conductance, trichome density). For leaf structure, we averaged values for three leaves (small, medium and
163 large sizes) per individual (leaf area, mass per area, thickness, density, dry matter content, saturated water
164 content and mass per area, petiole area to leaf area ratio, petiole mass to leaf mass ratio and petiole mass
165 per area), and for stem structure and allocation we measured one branch per individual (wood density and
166 Huber value). For leaf elemental and isotopic composition, we analyzed a mixture of 10-20 ground leaves
167 per individual (concentrations per leaf mass of carbon, nitrogen, phosphorus, potassium, calcium,
168 magnesium, iron, and trace metals, concentration per leaf area of nitrogen, and chlorophyll, and carbon:
169 nitrogen and nitrogen: phosphorus ratios, and carbon isotope discrimination). For plant hydraulic traits, we
170 measured two leaves per individual 3-5 individuals per species for turgor loss point, and 3-7 biomass-
171 dominant species from eight sites, we measured 15-48 leaves from 3-5 individuals per species for maximum
172 leaf hydraulic conductance and leaf water potential at 80% decline of conductance. For 3-12 biomass-
173 dominant species from six sites, we measured photosynthetic CO_2 response curves and calculated
174 photosynthetic maximum electron transport rate and maximum carboxylation rate traits for 3-27 leaves
175 from 1-12 individuals per species (see Table 2 for site abbreviations, and number of species per trait in
176 Table S3). Based on nutrient composition, isotope ratio and stomatal data, we derived additional modelled
177 trait values for time-integrated photosynthetic assimilation and stomatal conductance, and maximum
178 stomatal conductance: nitrogen concentration ratio. Additionally, we compiled maximum plant height and
179 seed mass from the TRY database[59].
180
181 ***Climate variables***
182 We focused on trait associations with species' native climate, using the mean climate variables of species'
183 native distributions based on the assumption that due to gene flow within a species' native range, average
184 trait values should correspond with average climatic conditions[6,60]. Additionally, we tested trait associations
185 with the climate at the sites at which species were sampled.

We estimated the climate of species' native ranges using occurrence data from the Global Biodiversity Information Facility[6,29,60–63]. Environmental variables were extracted and computed using R software (version 4.2.1[64]). Species occurrence data were obtained via the 'rgbif' R package [65], and filtered to retain herbarium records collected since 1950. Records lacking latitude or longitude, duplicates, and non-wild occurrences (such as botanical gardens or cultivated urban trees) were excluded [6,65,66]. Species data were further limited to observations from the United States, Canada, and Mexico, unless the species had a known global distribution. After screening with the *clean_coordinates* function of the 'CoordinateCleaner' package to remove points with quality issues (i.e, coordinates that represent administrative capitals, country centroids, equal or zero latitude and longitude, or a 1-degree radius around the GBIF headquarters), descriptive climate statistics were calculated at the species level. As our goal was to test trait–climate relationships, we opted to use direct occurrence points rather than species distribution maps based on ecological niche models to determine species' climatic envelopes to avoid bias in our analyses[6].

For the coordinates of each species' occurrences, and also for the coordinates of the study sites, four environmental variables were extracted from publicly available raster datasets, mean annual temperature and precipitation, (MAT and MAP, respectively; WorldClim[67]), and potential evapotranspiration and aridity index (MAPET and MAAI, respectively; CGIAR-CSI, NCAR-UCAR[68]. From the monthly data of these four variables, we calculated the growing season temperature (GST), growing season precipitation (GSP), growing season potential evapotranspiration (GSPET) and growing season aridity index (GSAI). The growing season was determined as the months with abundant soil moisture (mean precipitation in mm $\geq 2 \times$ mean temperature in degrees Celsius) and minimum temperature above 4ºC, the temperature below which water becomes too viscous to pass through membranes [69]. The raster layers were combined using the *'stack'* function from the 'raster' package[70], and environmental data for each occurrence point were extracted using the *extract* function from the 'dismo' package[67]. To characterize the climate of the sampling location, we extracted the same eight variables from 1×1 km grid cell around the centroid of each site (Table S2). While these datasets effectively capture broad-scale environmental trends, they are not suited for fine-scale variation such as microclimate factors—e.g., local temperature, light, moisture, or soil composition[71] (Complete dataset provided in Medeiros et al. *to be submitted*).

Statistical analyses

Statistical analyses and visualizations were conducted using R software (versions 3.4.4 and 4.0.2[64]) along with packages from the CRAN repository (detailed stepwise methods are found in https://github.com/NidhiVinod/Traits-and-Climate).

To establish the significance of species-level trait variation within and across sites, we performed two analyses of variance (ANOVA). First, we tested the variation across species in each trait, and second, we

used a nested analysis of variance to test for variation across sites, and variation among species within sites. Trait data were log-transformed to improve normality and homoscedasticity before the analysis. For traits containing zero or negative values, a constant of the absolute value of the minimum value plus 1 was added prior to transformation. We used packages car, MASS, dplyr, tidyverse, and TukeyC for this analysis.

To summarize the covariation of climate variables, we conducted a principal components analysis (PCA) using 'stats' package[64].

Given our interest in the absolute predictive and explanatory power of climate variables, rather than evolutionary processes, we applied ahistoric (nonphylogenetic) analyses. We applied four approaches to test the numbers and strength of associations of traits with the eight climatic variables (GSP, GST, GSAI, GSPET, MAP, MAT, MAAI, MAPET). First, for each trait-climate association we conducted Spearman rank correlations and Pearson correlations for both untransformed and log-transformed data, and considered the highest of the three *r*-values ("highest *r*-value"). We then determined the proportion of trait-climate associations that was significant for each climate variable. To assess whether these proportions differed between climate variables, we conducted pairwise proportion tests using the *prop.test* function in R versions 3.4.4 and 4.0.2[64]. Second, for each climate variable, we conducted analyses of variance to test for differences among the climate variables in their highest *r*-values. Third, we tested for differences among each pair of climate variables in the strength of all trait correlations (including both significant and nonsignificant) using a meta-analysis of the highest *r*-values[11] adapted from *MetaWin*[72] using the package 'metafor'[73]. Finally, we tested the relative importance of MAT and MAP in co-determining traits, by fitting multiple regression models that included MAT and MAP as independent variables and functional traits as dependent variables; we then performed a hierarchical partitioning analysis using the 'hier.part' package to calculate the percentage contribution of each variable to the prediction of traits[74]. These models were fitted to untransformed and log-transformed data, and we selected the highest R2 based between log-transformed and untransformed data for each trait. We calculated the percentage of traits that were best explained by MAT or MAP, and the mean percent contribution of MAT and MAP to the determination of each trait. Each of the above tests were conducted across all the species-site combinations, first considering trait associations with the mean climate of individual species' native ranges, and second, considering associations with the climate of species' sampling sites.

We tested the relative variation in the climate variables of species' native ranges and of the study sites using coefficients of variation for log-transformed data after applying a constant of the absolute value of the minimum value plus 1, and tested for differences among these using standardized likelihood ratio test (SLRT[75]).

253 **Results**

254 We found strong variation across species for vast majority of the traits (ANOVAs; $P < 0.001$ to $P < 0.05$
255 for most traits; Table S4). The majority of variance, on average 92%, was explained by species-differences
256 (ANOVAs; $P < 0.001$ to $P < 0.05$; Table S4). Considering species nested within site, 35% of variation was
257 explained by site differences, 61% by species within sites, and 4% by variation among individuals within
258 species (Table S4).

259 We conducted a principal components analysis (PCA) to clarify the pattern of variation in the mean
260 climate variables of species' native distributions (Fig. 1c). The first two axes, Climate-PC1 and Climate-
261 PC2, accounted for 60.4% and 21.8% of the variation, respectively, and 82% in total (Table S5). Climate-
262 PC1 was related to the aridity gradient from cool, wetter sites to hotter, drier sites, i.e., from low to high
263 MAT, MAPET and MAAI, and from high to low GSP and MAP. Climate-PC2 captured orthogonal
264 variation in climatic warmth and evaporative demand, i.e., MAT and MAPET (Table S5, Fig 1). Both
265 PCA axes were related to latitude, though stronger for Climate-PC2 than Climate-PC1 ($r = 0.86$ and 0.32,
266 respectively; $P < 0.001$; Table S5; Fig. S1). In the comparison of the relative variation in MAP and MAT
267 across species' mean climates and site mean climates, MAT showed the stronger variation, with
268 coefficients of variation ± standard error of 0.145±0.0080 and 0.175± 0.00973 respectively, relative to
269 0.00792±0.0044 and 0.00923± 0.00511 for MAP (Table S16, Table S17).

270 Across species, 58 of the 59 measured plant traits were related to one or more climate variables
271 representing the mean of species' climate distributions, all but foliar carbon: nitrogen ratio (Fig. 3; Tables
272 S7 and S8). Similarly, when considering the mean climate of species' sampling sites, 57 of 59 measured
273 traits were related to one or more climate variables, all but leaf dry matter content and bark thickness to
274 stem diameter ratio (Table S9 and S10).

275 Overall, traits were more strongly correlated with precipitation than with temperature. The overall
276 findings for stronger relationships with climatic precipitation than temperature at continental scale is
277 illustrated with five example traits, wood density (WD), turgor loss point (TLP), carbon isotope
278 discrimination ($\Delta^{13}C$), leaf area (LA), foliar nitrogen content per mass (N_{mass}), and leaf mass per area (LMA)
279 (Fig. 2). Thus, TLP, $\Delta^{13}C$, N_{mass} and LMA, had stronger associations with GSP than GST, and TLP, $\Delta^{13}C$,
280 LA, N_{mass} and LMA had stronger associations with MAP than MAT.

281 In our analyses across all the traits, considering associations with the mean climate variables for
282 species' native ranges, a greater proportion of trait correlations were significant for MAP versus MAT
283 (proportion tests; Table 3). Further, trait-climate correlations were stronger for MAP versus MAT when
284 considering only significant correlations (ANOVAs; Fig. 3; Table S11), and for MAP versus MAT and also
285 for GSP versus GST, when considering all their correlations regardless of significance (meta-analysis;
286 Table 3). Further, multiple regression analyses of each trait with respect to MAT and MAP also supported

the greater importance of MAP (Table S12). Across the 59 traits, the multiple regressions showed a mean ± standard error of the adjusted r^2 of 0.14 ± 0.015; 84% of traits showed a higher contribution of MAP, and 15% of MAT, and across all the traits, the mean ± standard error of the percent contribution of MAP was 77.6 ± 3.23% whereas that of MAT was 22.3 ± 3.23%.

Trait variation was also strongly explained by climatic variables beyond temperature and precipitation, i.e., potential evapotranspiration and aridity index. Indeed, MAPET and MAAI both had higher proportions of significant trait correlations and stronger trait correlations than MAT (Table 3; Fig. 3). Considering growing season variables, GSPET and GSAI showed significantly stronger overall trait correlations than GST (Table 3).

Comparing growing season and annual climate variables, for temperature, the growing season mean had a greater proportion of significant trait correlations and stronger correlations overall than the annual mean (Table 3; Fig. 3), and for aridity index, the growing season mean had stronger overall trait correlations and stronger significant correlations than the annual mean (Table 3; Fig. 3; Table S11). For precipitation and potential evapotranspiration, growing season and annual means did not differ in the proportion or strength of correlations (Table 3; Fig. 3; Table S11).

The findings were similar when considering trait associations with the climate of species' sampling sites, rather than the mean climates of species' native ranges. Thus, a greater proportion of trait correlations were significant, and overall correlations, and significant correlations were stronger, for MAP versus MAT (Table S13 and S14; Fig. S2).Further, MAAI and MAPET both had a higher proportion of significant trait correlations, and stronger associations than MAT, whether considering all correlations or only significant ones (Table S13 and S14; Fig. S3). Considering growing season variables, GST was statistically similar in the proportion and strengths of significant and overall trait correlations as GSP, GSPET and GSAI (Table S13 and S14; Fig. S3). Multiple regression analyses of each trait with respect to MAT and MAP of the climate of species' sampling sites also supported the greater importance of MAP (Table S15). Across the 59 traits, the multiple regressions showed a mean ± standard error of the adjusted r^2 of 0.12 ± 0.016; 78.0% of traits showed a higher contribution of MAP, and 22% of MAT. Across all the traits, the mean ± standard error of the percent contribution of MAP was 79.7 ± 3.64% whereas that of MAT was 20.3 ± 3.64% (Table S10). Comparing growing season and annual climate variables, GST had a higher proportion of trait correlations, and stronger overall and significant correlations than MAT. For aridity index, the annual mean had a higher proportion of trait correlations, and stronger overall and significant correlations than the growing season mean (Tables S13 and S14; Fig. S3). For precipitation and potential evapotranspiration, growing season and annual means did not differ in the proportions or strength of correlations (Table S13 and S14; Fig. S3).

Discussion

Switching of patterns from global scale to the continental scale across the contiguous USA and northern Mexico

The important role of both precipitation and temperature in predicting traits across the contiguous USA and northern Mexico are consistent with several previous studies at continental and global scales[39,43,49,52]. Yet, we found a strongly novel finding with MAP being a better predictor than MAT, contrasting with the pattern shown in global studies, in which temperature is a better predictor of traits than precipitation[11,38,76,77]. Indeed, across all our analyses of continental-scale trait-climate associations, we found that while traits have substantial associations with temperature, precipitation, potential evapotranspiration and aridity index, precipitation is a stronger predictor of plant traits than temperature. Consistent with our findings, a recent continental analysis for over 7 types of ecosystems spanning from arid, tropical to temperate in Australia highlighted the importance of MAP, though without a comparison of the relative strengths of trait associations with MAT vs. MAP[52].

The difference in the findings of continental from global tests of trait-climate associations would emerge for multiple reasons associated with the disparity in scales (Table 1). At first sight, one might assume that the relative strengths of trait-climate associations may reflect the variation in specific climate variables, and thus that the stronger trait associations with temperature than precipitation at global scale corresponds to the greater variation in temperature, whereas the stronger trait associations with precipitation at continental scale reflects a greater variation in precipitation than temperature. However, in our study, and in a global-scale trait-climate study[11], MAT exhibited greater climatic variation compared to MAP. The stronger role of MAP than MAT as a driver of trait variation continentally may rather be linked with the associations of climate traits. Globally, the dominant climatic gradient is latitudinal, with variation from colder, drier climates at high latitudes to warmer, wetter climates in the tropics. By contrast, for the continental gradient in our study, a climatic aridity gradient is dominant, from cooler, moister climates to drier, warmer climates, due to continentality and elevational variation[78]; such a gradient would reinforce the importance of moisture, as water stress in combination with high temperature represents a stronger compound stress. Indeed, in our principal components analysis of climate variation across the species' native ranges, temperature, and precipitation partly inter-related, with Climate-PC1 showing a strong aridity gradient, with temperature and precipitation negatively correlated, and Climate-PC2 showed additional variation in temperature, showing the influence of the latitudinal gradient, here independent of precipitation. The importance of climatic rainfall as a trait driver is also consistent with its influence on climatic temperature at regional and continental scales; for example, reduced water levels can contribute to high growing season climatic temperatures[79–81]. We note that stronger trait-climate associations would also arise continentally than globally given the relative floristic similarity by contrast with a global analysis[52], though

355 potentially weaker relationships might be expected given the more constrained species variation[11]. Notably,
356 the global-scale study by Moles et al. 2014 focused on all life forms—including herbaceous and woody
357 species—our study focused on woody species. Microclimate, specifically MAT and MAP experienced by
358 understory herbaceous species varies drastically from woody canopy species[82], and thus trait relationships
359 with macroclimate variables may be weaker for herbaceous than woody species, especially as the life cycle
360 and phenology of herbaceous species might reduce the time of year in which they are present and/or active,
361 altering trait-climate relationships[61,83]. We conclude from the strong divergence of our continental study
362 from global studies of the priorities of trait-climate associations that the global pattern is not generalizable
363 across continents. However, we also note that the specific pattern found across the sites in our study, may
364 differ from that of other continents, depending on the floristic composition of the continent, the specific
365 ranges of climate variables, and their inter-relationships. Importantly, the trends observed continentally may
366 be expected to be more potentially actionable, relative to global trends, which combine disparate
367 ecosystems with similar climates[11].
368 Another novel aspect of our study was the test of whether growing season climate variables would
369 be stronger predictors of traits than annual variables. A greater importance of growing season climate would
370 be expected in considering that in temperate regions the leaf expansion and development, and the bulk of
371 photosynthetic activity occurs during the growing season[84]. Indeed, with GST had a greater proportion of
372 significant trait correlations and stronger correlations than MAT, likely because temperature directly
373 constrains developmental and physiological processes such as leaf expansion, nutrient assimilation and
374 photosynthesis during periods of active growth. However, for precipitation and potential
375 evapotranspiration, annual and growing-season variables showed similar proportions of significant trait
376 associations and comparable association strengths, and for aridity index, the annual mean showed stronger
377 correlations than the growing season mean (Table 3 and Tables S11, S13 and S14). These findings are
378 consistent with the pronounced importance of precipitation as a driver of trait associations across our
379 continental gradient. For woody species, given deep soils, the availability of moisture in the growing season
380 may depend on precipitation falling year-round, with water stored from non-growing season periods
381 remaining accessible during the growing season, buffering plants against short-term drought. Additionally,
382 traits associated with survival during the non-growing season—such as cold tolerance, drought resistance
383 during late summer, or maintenance of perennial structures—may be shaped by climatic conditions across
384 the entire year, reducing the contrast between growing season and annual variables as separable predictors
385 for MAP. Similarly, aridity index would reflect cumulative water balance and atmospheric demand across
386 the entire year, such that plant traits respond more strongly to annual than to growing-season aridity (Table
387 1).
388

389 ***Mechanistic associations of traits with precipitation and aridity gradients***
390 Our findings of precipitation as the strongest driver of plant traits reflects multiple functional trait
391 adaptations along a continental aridity gradient, that would confer stress resistance and/or stress avoidance
392 (Table S1). Species associated with lower precipitation levels have a range of traits shown in previous
393 studies to be adaptive, including shorter stature, smaller cell sizes, thicker leaves with higher leaf mass per
394 area, and leaf dry mater content, smaller leaf size, denser trichomes lower stomatal density, lower TLP,
395 lower $\Delta^{13}C$, higher wood density, higher Huber value, lower saturated water content, lower hydraulic
396 conductance, safer hydraulic thresholds and lower nitrogen mass per area compared to plants adapted to
397 moister regions (Fig. 2; Table S7). These examples of the associations of traits across many plant levels of
398 organization and functional roles highlight the extraordinary importance of precipitation, and aridity, as
399 important predictors of plant traits at continental scale (Fig. 2, Table S7).
400
401 ***Limitations of the analyses***
402 Our study highlights important relationships among traits and climate variables, and importantly expands
403 on previous studies in the number of traits measured according to standard protocols for a diversity of
404 ecosystems included for trait-climate associations at continental scale. Our analysis was based on
405 standardized field-collected data for 59 traits across 16 ecosystems, capturing diverse woody species across
406 ecosystems at the continental scale. Notably, previous large-scale trait–climate studies have often relied on
407 databases that include non-standardized trait measurements, often compiled for different plants of each
408 species growing in disparate locations.
409 However, our analysis requires key caveats, pointing to avenues for future studies to achieve yet
410 higher resolution. Our climate data were modelled from species occurrences, a widely used approach,
411 though relatively coarse, and which may limit the precision of trait-climate relations; finer-resolution
412 climate data could enhance the relationships observed in our study. Indeed, microclimatic temperature often
413 differs from regional weather stations, and modelled macroclimatic temperature [85]. Further, climatic
414 moisture variables may have a complex relationship to species' water availability in the field. The
415 availability of water to plants depends on a suite of factors, including the seasonal distribution of rainfall,
416 hydrology, soil depth, soil type (including soil moisture-holding capacity and the soil moisture
417 characteristic curve), access to groundwater and on temperature, which determines both whether the
418 precipitation falls as rain or snow and the level of evaporative demand[44,86–88]. Notably, the growing season
419 variables are estimated by thresholds, therefore it is likely may not necessarily correspond to actual ranges
420 of water availability experienced by the plants in their growing seasons. Additionally, our use of aggregated
421 climate variables (e.g., mean annual or growing season values) may mask the effects of extreme events or
422 intra-seasonal variability—factors that can strongly influence trait adaption and assembly. While we tested

for differences between growing season and annual climate variables, the temporal resolution of the available data constrains our ability to evaluate carry-over effects from previous years or to separate the impacts of long-term climate trends from short-term anomalies.

Our finding of stronger trait associations with precipitation than temperature, in contrast to global studies, emphasizes the importance of considering scale and specific systems in examining the priorities of trait-climate associations. That point remains robust, though, as noted above, the strength of given climatic variables in driving trat associations would vary across other continents that differ in the ranges of climate variables and their inter-relationships and in their floras. Further, the designation and sampling of species within continents would also play a role in the specific trait-climate associations discerned. Thus, despite our including a wide range of species and representative ecosystems from the contiguous USA and northern Mexico, a wider definition and greater sampling across of the North American continent, including Canada and Mexico would likely shift the specific findings for relative trait associations to a degree. However, our finding of a stronger role for precipitation than temperature in our studied continental gradient remains a robust demonstration of an important departure from global patterns.

Overall conclusions and practical applications

Our findings highlight the ecological importance of multiple climate variables, and in particular, of precipitation as a stronger trait predictor than temperature in shaping plant functional strategies continentally, differing from the findings globally where temperature can more strongly predict species traits. Precipitation as a driver of traits continentally would influence species occurrences and distributions. As droughts become more frequent and intense, species may face limits to their adaptive capacity associated with their traits, restricting their ability to persist outside certain precipitation ranges. The importance of precipitation continentally points to the need for future studies to analyze continental climate-trait patterns across the globe. Models projecting species and trait distribution for future climate scenarios should incorporate precipitation as a driver of traits regionally and continentally rather than relying primarily on temperature, though often modelled scenarios show greater resolution of potential shifts in temperature compared to precipitation[89,90]. Precipitation-driven variation in species and traits also impacts ecosystem scale functions such as ecosystem carbon and water fluxes, since traits can directly impact ecosystem productivity and fluxes[91] with feedbacks on future precipitation regimes. Our study suggests that the consideration of the wide range of trait-climate associations will improve predictions of across scales, from species-distributions to the terrestrial climate cycle, to future climate, and locally, improve restoration efforts through incorporating species with drought tolerance in regions prone to droughts and heatwaves.

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Author Contributions

Conceptualization of the project: LS, CM, NV, MD, ATT, NK; writing of original draft: NV and LS; formal analyses: NV, CM, KCC and LS; investigation- species sampling: CM, NV, AO, MB, MD, HD, JZ, CH, ST, LS; investigation- collection of trait data: CM, NV, AO, MB, MD, HD, LS; investigation- site logistics and species abundance: LS, JF, NMH, CM, SM, GPJ, DJ, JAL, RMA, JDW, KAT, MD, ATT, NK; data visualization: NV and CM; review & editing: all.

Conflict of Interest

None.

Data and Materials Availability

All code used for this analysis is available on GitHub Trait-Climate repository:
https://github.com/NidhiVinod/Traits-and-Climate

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728
729 **Table 1.** Contrasts in elements of pattern and processes between global and continental scale gradients that would influence the priorities of trait-
730 climate association across species, and their generalizability for prediction.

Pattern or process element	Global Scale	Continental Scale	Potential influence on across-species trait-climate associations
Concurrence of temperature and precipitation gradients	The dominant climatic gradient is latitudinal, with variation from colder, drier climates (high latitudes) to warmer, wetter climates (tropical). [92–95]	Climatic aridity gradients are often found, from cooler, moister climates to drier, warmer climates, due to continentality and elevational variation [78]	The global latitudinal gradient aligns adaptation to low/high moisture with low/high temperature, whereas aridity gradients as typical at continental scale emphasize the importance of moisture, as water stress is stronger when combined with high temperatures
Climatic overlap	Similar climates can exist in extremely floristically distinct regions (e.g., Mediterranean vs. Australian shrublands) [96]	Floristics more cohesive typically for given climates within a continent [97,98]	Trait-climate associations may be stronger continentally than globally
Biogeographic context	Strong floristic/evolutionary differences across the spatial extent (Erwin 2009)	Stronger influence of shared evolutionary and biogeographic history [97]	Trait-climate associations may be stronger continentally than globally
Species pool	Higher species variation due to global species turnover and greater phylogenetic heterogeneity [99]	More constrained species variation due to consistent regional species pools and lower phylogenetic heterogeneity [100].	Trait-climate associations may be stronger globally than continentally
Generalizability of trait-climate associations	May obscure local/regional patterns (Towers et al. 2024)	Captures variation without overgeneralization (Towers et al. 2024)	Trait-climate associations may be more generalizable continentally than globally, and thus more actionable and policy-relevant (e.g., national/continental planning)

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Table 2. Study sites across the continental USA and northern Mexico, ordered from West to East with site codes (Fig. 1). Sites with forest type, number of species, growing season temperature (GST), growing season precipitation (GSP), growing season potential evapotranspiration (GSPET), growing season aridity (GSAI), mean annual temperature (MAT), mean annual potential evapotranspiration (MAPET), and mean annual aridity (MAAI).

Site	Vegetation type	# sp.	GST	GSP	GSPET	GSAI	MAT	MAP	MAPET	MAAI
Sweeney Granite Mountains Desert Research Center (SGMD)	Desert	28	8.68	122	542	0.228	16.6	263	2738	0.095
Stunt Ranch Santa Monica Mountains Reserve (SRSM)	Chaparral	26	12.9	392	734	0.563	16.4	412	1926	0.213
Centro de Investigación Científica y de Educación Superior de Ensenada (ENSE)	Coastal sage scrub	22	13.0	183	408	0.336	16.4	256	1588	0.121
Yosemite Forest Dynamics Plot (YFDP)	Montane wet forest	20	7.87	605	720	1.09	10.7	977	1812	0.539
San Joaquin Experimental Range (SJER)	Shrubland, evergreen forest	14	10.3	494	569	1.03	16.6	552	2118	0.260
Onion Creek (OC)	Mixed riparian woodland	19	9.75	137	441	0.330	6.46	1122	1486	0.754
Angelo Coast Range Reserve (ACRR)	Mixed evergreen-deciduous forest	21	8.85	1557	641	3.30	11.4	1613	1365	1.18
Wind River Forest Dynamics Plot (WFDP)	Evergreen forest	18	10.8	1086	820	1.96	9.05	2005	1081	1.85
Missouri Ozark AmeriFlux (MOFL)	Deciduous forest	20	16.6	858	1239	0.753	12.4	1013	1374	0.737
Niwot Ridge Mountain Research Station (NIWO)	Evergreen forest	12	8.58	218	616	0.352	0.039	668	1202	0.555
Harvard Forest (HARV)	Mixed evergreen-deciduous forest	26	13.8	656	809	0.852	7.01	1105	1004	1.10
Rutgers University Pinelands Field Station (RPFS)	Atlantic coastal pine barrens	18	15.3	895	1088	0.897	11.7	1159	1237	0.937
Smithsonian Conservation Biology Institute (SCBI)	Mixed evergreen-deciduous forest	12	15.1	828	1080	0.825	11.5	1038	1232	0.841
Smithsonian Environmental Research Center (SERC)	Deciduous forest	22	16.7	860	1189	0.768	13.2	1103	1356	0.813
Great Smoky Mountains National Park (GRSM)	Deciduous forest	26	14.6	1163	1175	1.08	12.7	1392	1295	1.07
Ordway Swisher Biological Station (OSBS)	Woody evergreen forest	24	20.5	1297	1625	0.802	20.5	1297	1632	0.794

Table 3. Tests of differences in the numbers and strengths of correlations and of 59 traits with mean climate variables for 328 species' site combinations across 16 ecosystems of the continental USA and northern Mexico. For tests of proportions of significant correlations, pairwise proportion tests were used; tests of correlations strength were based on meta-analysis (log-ratio tests); Significant differences at P < 0.05 are highlighted in bold and with an asterisk for P<0.05. Number of asterisks follow standard convention of p-levels (* <0.05; ** <0.01; *** <0.001). Rows are ordered within sections from the highest to lowest proportion of significant trait correlations

Climate Variable 1	Climate Variable 2	Proportion of significant trait correlations		*P*-value (proportion tests)	Meta-analysis log ratio of variable 1 / variable 2 (95% confidence intervals)	*P*-value (meta-analyses)
		Variable 1	Variable 2			
Annual climate variables						
MAP	MAT	**0.881**	**0.492**	**<0.001**	**1.37 (0.75, 1.367)**	**<0.001**
MAAI	MAT	**0.831**	**0.492**	**<0.001**	**1.39 (0.858, 1.928)**	**<0.001**
MAAI	MAP	0.831	0.881	0.70	0.015 (0.2, 0.238)	0.89
MAPET	MAAI	0.712	0.831	0.094	0.242 (0.031, 0.514)	0.082
MAPET	MAP	0.712	0.881	0.98	0.282 (-0.665, 0.101)	0.15
MAPET	MAT	**0.712**	**0.492**	**0.012**	**1.18 (0.634, 1.183)**	**<0.001**
Growing season climate variables						
GSP	GSPET	0.797	0.746	0.33	**0.163 (0.064, 0.262)**	**0.0012**
GSP	GST	0.797	0.678	0.10	**0.835 (0.452, 1.219)**	**<0.001**
GSP	GSAI	0.797	0.695	0.14	**0.914 (0.577, 1.251)**	**<0.001**
GSPET	GSAI	0.746	0.695	0.24	**0.735 (0.362, 1.107)**	**<0.001**
GSPET	GST	0.746	0.678	0.73	**0.651 (0.283, 1.018)**	**<0.001**
GST	GSAI	0.678	0.695	0.50	0.052 (-0.523, 0.627)	0.86
Annual vs growing season climate variables						
MAAI	GSAI	0.831	0.695	0.94	**0.851 (0.463,1.239)**	**<0.001**
MAP	GSP	0.797	0.881	0.84	-0.048 (0.411, 0.316)	0.80
MAPET	GSPET	0.712	0.746	0.42	0.069 (-0.276, 0.414)	0.70
GST	MAT	**0.678**	**0.492**	**0.031***	**0.770 (0.151,1.389)**	**0.015**

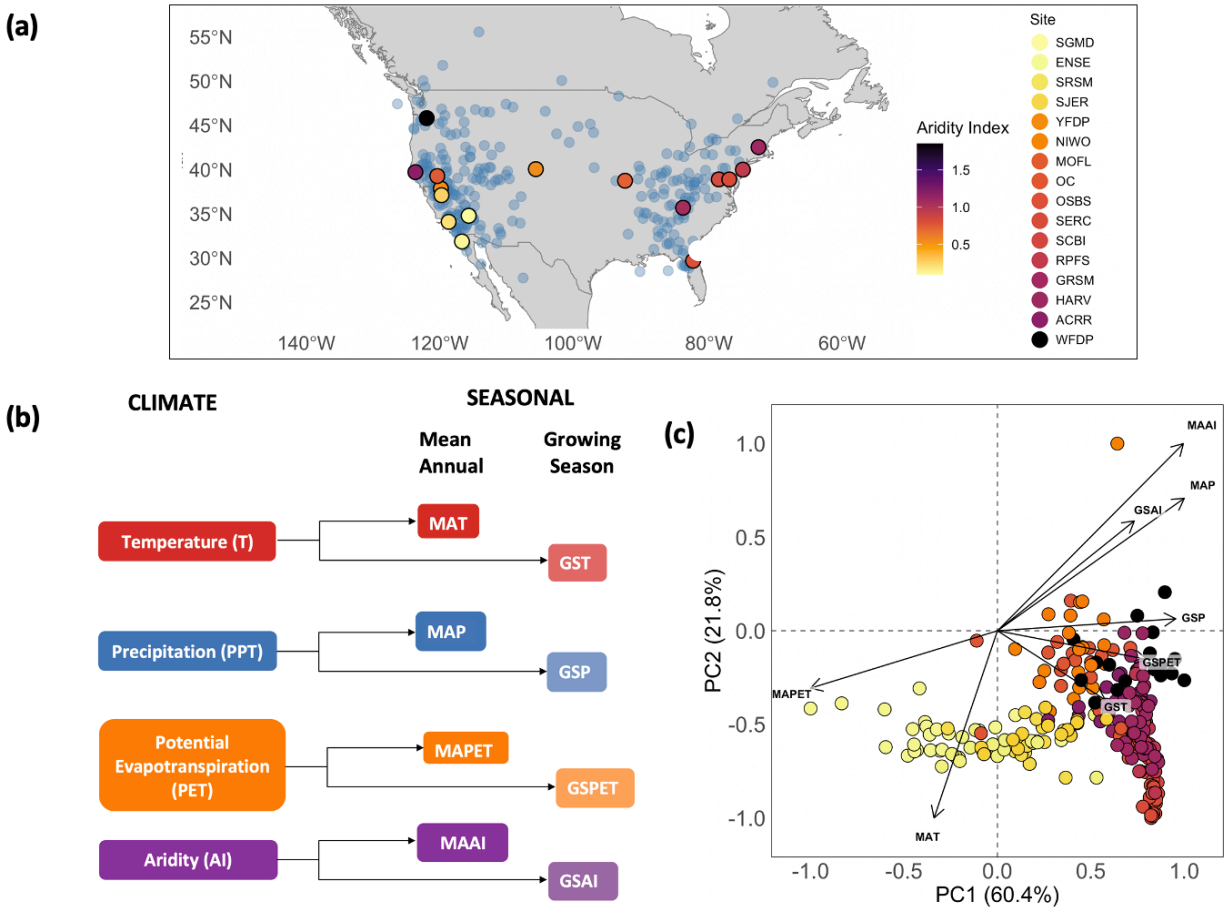

Figure 1. (a) Mean coordinates of the species' native distributions for 328 species-site combinations for 246 species sampled from 16 ecosystems 246 unique species in this study and the 16 sites sampled across the USA and northern Mexico (see Table 2 for site codes) (b) Flowchart of climatic variables included in this study, i.e., mean annual precipitation (MAP), temperature (MAT), potential evapotranspiration (MAPET), and aridity (MAAI) and growing season mean precipitation (GSP), temperature (GST), potential evapotranspiration (GSPET), and aridity (GSAI). (c) Principal components analysis of the climate variables; the points represent the mean climate of species' native distributions, and are colored according to the species' sampling locations.

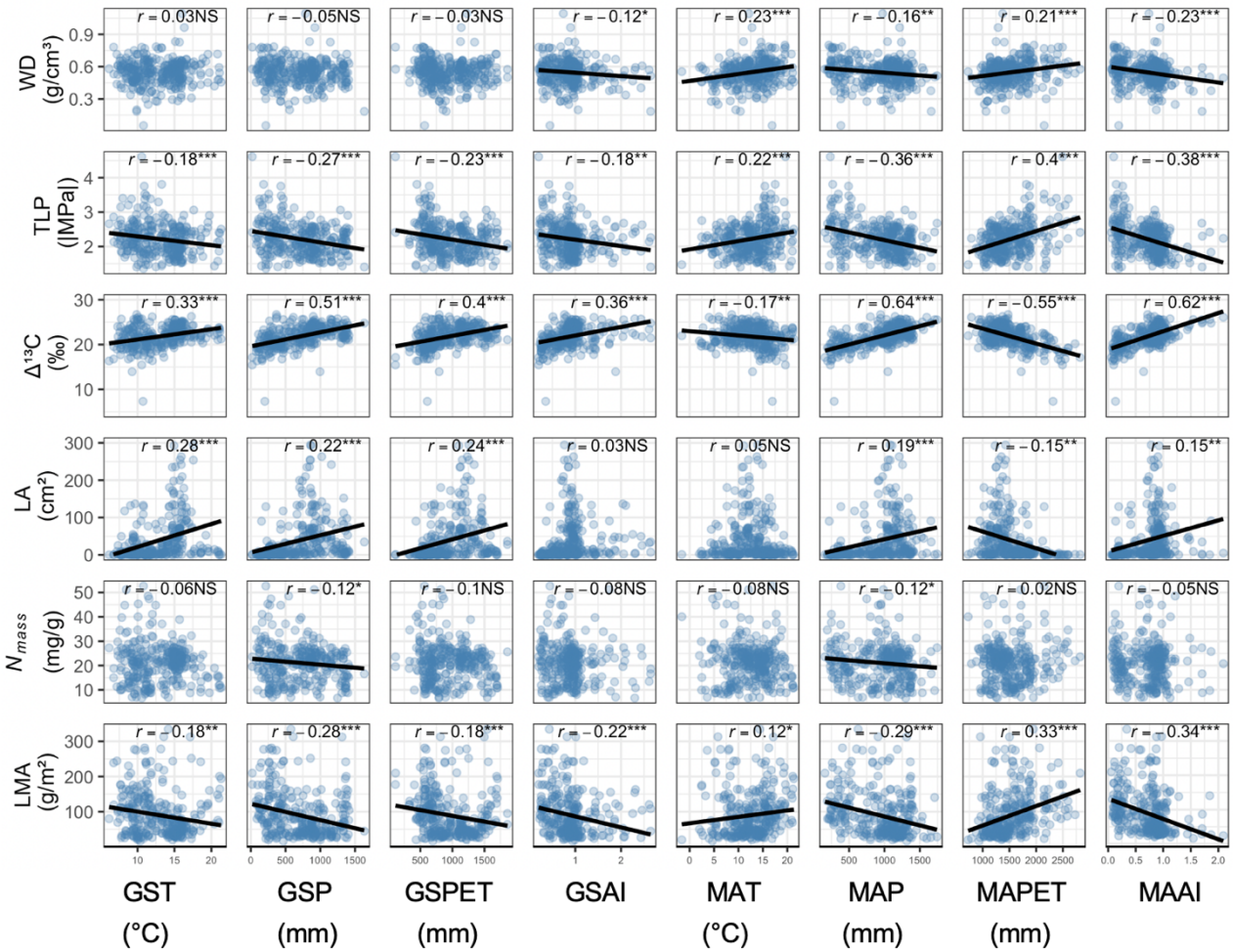

Figure 2. Illustrative relationships of traits with the mean climate of species' native distributions for 328 species-site combinations sampled from 16 ecosystems of the continental USA and northern Mexico. : relationships of wood density (WD), turgor loss point (TLP), carbon isotope discrimination ($\Delta^{13}C$), leaf area (LA), foliar nitrogen content per mass (N_{mass}), leaf mass per area (LMA), with growing season temperature (GST), growing season precipitation (GSP), growing season potential evapotranspiration (GSPET), growing season aridity index (GSAI), mean annual temperature (MAT), mean annual precipitation (MAP), mean annual potential evapotranspiration (MAP), mean annual aridity index (MAAI).

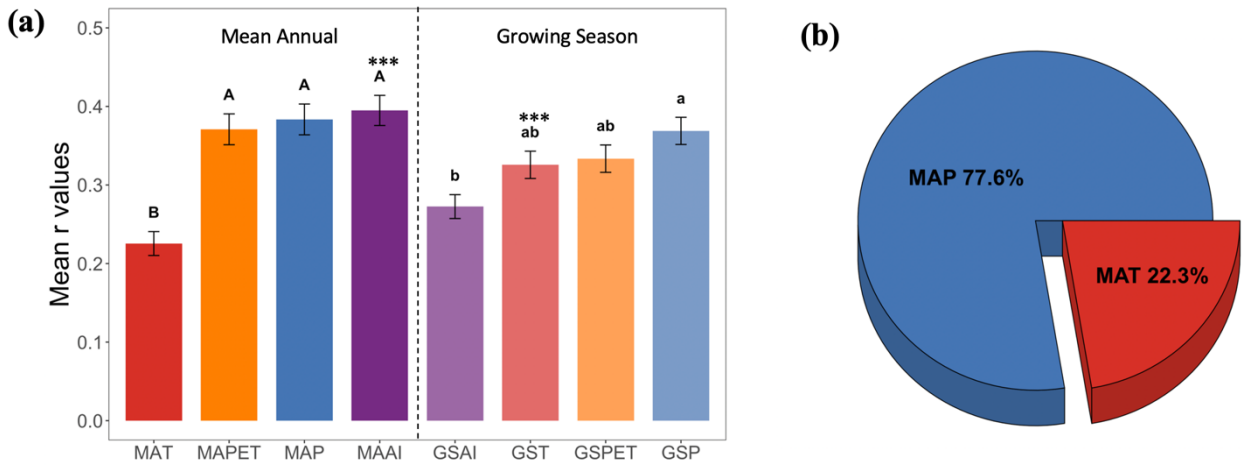

Figure 3. Mean correlation coefficients (*r* values) ± standard error) for significant associations of traits with the mean climate of species' native distributions, across 328 species' site combinations sampled from 16 ecosystems of the continental USA and northern Mexico. Mean annual and growing season climate variables for: temperature (MAT and GST, respectively), precipitation (MAP and GSP), potential evapotranspiration (MAPET and GSPET), and aridity (MAAI and GSAI). Letters above each bar indicate significant differences among mean annual or growing season mean variables based on Tukey comparisons, such that; bars with different letters within mean annual or growing season mean variables are significantly different at $P < 0.05$ (Table S11). Asterisks above the bars indicate significant differences between growing season and mean annual values (ANOVAs; $p<0.05$; Table S11).